Exercise 2 Aggregate Functions And Grouping

1. COUNT () Function

```
SELECT (COUNT) ID AS Number_of_employees
FROM employee;
```

Outcome 10 employees

2. SUM() Function

```
SELECT First_name, Last_name, Department,
        (SUM) AS total_salary,
FROM employees
WHERE salary = 'HR department';
```

ID	First_name	Last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55 000	2018-06-15	New York
4	Sarah	Brown	IT	53 000	2021-05-25	New York
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

3. AVG Function

```
SELECT First_name, Last_name, department, Salary,
        (AVG) AS total_salary
FROM Employees
WHERE salary = 'HR department';
```

ID	First_name	last_name	Department	Salary	Hire_date	City
2	Jane	Smith	HR	48 000		
8	Jessica	Moore	HR	51 000		

4. MIN() and MAX() Functions

```
SELECT First_name, Last_name, Salary,
    (MIN) AS lowest_salary,
    (MAX) AS Highest_salary
FROM employees;
```

ID	First_name	Last_Name	Salary	High & low Salary
6	Emily	Davis	62 000	Highest Salary
2	Jane	Smith	45 000	Lowest Salary

5. Group By Statement

```
SELECT DISTINCT department,
                Salary
FROM employees_db
GROUP BY department;
```

ID	Department	Salary
	IT	220 000
	HR	99 000
	Finance	119 000
	Marketing	105 000

6. GROUP BY and COUNT()

```
SELECT (COUNT city) AS location
FROM employees_db
GROUP BY cities;
```

City	Number Of Employees
New York	2
Chicago	3
Los Angeles	2

7. SELECT department, salary
(SALARY AVG) AS department
FROM employees_db
ORDER BY salary DESC;

Department	Salary
IT	220 000
Finance	119 000
Marketing	105 000
HR	99 000

8. HAVING Statement

SELECT DISTINCT departments, salary
FROM employees_db
WHERE departments
HAVING salary > 100 000;

Departments	Salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING and ORDER BY

SELECT DISTINCT first_name, last_name, city
FROM employees_db
WHERE city = 'Chicago'
ORDER BY Number_of_employees DESC;

City	Number_of_employees
Chicago	3
New York	2
Los Angeles	2